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Advanced Azimuthal Resistivity Analysis for Better Understanding of Subtle Carbonate Oil Reservoirs

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Abstract

This paper presents a novel approach to processing azimuthal resistivity measurements, which significantly enhances the detection and characterization of thin porosity zones within dense limestone formations. The deterministic method utilizes Newton non-linear inversion techniques to convert electromagnetic data obtained during drilling operations into comprehensive multi-layer resistivity models.

In contrast to traditional distance-to-boundary methods that often encounter difficulties in low-resistivity contrast environments, the algorithm is specifically designed to excel in these challenging conditions. Under optimal circumstances, this technique can achieve investigation depths exceeding 40 feet from the wellbore while simultaneously resolving formation thickness and resistivity values across multiple layers.

The framework is highly adaptable, accommodating a range of layer configurations, from straightforward two-layer models to intricate ten-layer structures, allowing it to address various geological complexities effectively. A key feature of the approach is the integrated misfit analysis, which provides thorough quality control by automatically identifying solution confidence and potential ambiguities.

The algorithm maintains a consistently high number of degrees of freedom, with user-controlled constraints that ensure reliable and robust solutions, even in scenarios characterized by minimal resistivity contrast.

A field examples are presented that illustrate how this approach successfully tracked and determined adjacent layers in a carbonate reservoir. This capability is crucial for developing strategies to navigate between porous layers. In instances where reservoir contact was lost due to faulting, the method effectively assessed the structural position and guided the optimal path for re-establishing access to the target zone.

In summary, the multi-layer azimuthal resistivity inversion technique introduced in this paper represents an advancement in the characterization of carbonate reservoirs, offering a reliable and effective solution for complex geological challenges.

How the Method Works

Drilling operations in complex geological environments require advanced techniques for interpreting underground rock layers through their electrical properties. A systematic approach based on the deterministic Newton's mathematical optimization method has been developed to interpret azimuthal

resistivity measurements in geologically challenging environments, particularly those characterized by relatively low resistivity contrasts and complex structural configurations.

The optimization process minimizes the misfit between electromagnetic tool measurements and theoretical model predictions through iterative refinement. Starting with an initial estimate of the resistivity structure, typically derived from offset wells data or regional geological models, the algorithm iteratively refines this model through mathematical optimization. During each iteration, the Jacobian matrix determines measurement sensitivity to changes in rock properties, essentially mapping how variations in layer thickness or resistivity affect the tool response at different frequencies and spacings. This sensitivity information guides parameter updates for subsequent iterations, with step sizes controlled by regularization parameters to ensure stable convergence toward accurate subsurface representation.

Layer thickness and resistivity values are treated as independent parameters requiring simultaneous optimization within the inversion framework. Forward modeling calculates the expected electromagnetic response for any given layered earth structure using Maxwell's equations, accounting for wave propagation through heterogeneous media, frequency-dependent attenuation, and phase shifts at layer boundaries. The electromagnetic field distribution depends on the contrast between layer properties, tool configuration, and operating frequency, with higher frequencies providing better vertical resolution but reduced depth of investigation. The inversion engine guided by constant user control of the parameters possible ranges systematically adjusts them to minimize differences between measured and modeled responses using a gradient-based optimization approach. The framework handles configurations ranging from simple two-layer cases to complex ten-layer arrangements, addressing various geological scenarios encountered in carbonate environments. Built-in quality assessment automatically evaluates confidence levels and identifies potential uncertainties arising from different type of noise, and final parameters correlation results.

User-defined constraints ensure reliable results, particularly when resistivity contrasts between layers are small. These constraints include physical bounds on layer thickness based on geological knowledge, and resistivity range limitations derived from offset well data. The constraint implementation uses penalty functions that prevent the algorithm from exploring geologically unreasonable parameter combinations while maintaining sufficient flexibility to honor the measured data. Newton-based methods demonstrate efficiency simplifying implementation and validation under field conditions where rapid decision-making is essential.

Practical deployment considerations favor systematic methods for field operations where reliability and speed are paramount. Statistical approaches require connection to specialized computing centers or cloud infrastructure for real-time performance, introducing dependencies on network stability and latency. In contrast, systematic methods operate effectively on standard drilling-site equipment without high-speed internet connections or parallel processing capabilities. This independence from external computational resources ensures reliable operation in remote locations, offshore platforms, or desert environments where connectivity may be limited or unreliable.

The uncertainty estimates from the systematic approach are calculated directly from how well the data fits and how well the model resolves features, providing immediate, data-driven confidence ranges without requiring prior statistical assumptions or parameter adjustments. When electromagnetic data shows good signal quality, the systematic solution proves to be adequate while avoiding the computational burden of exploring unlikely parameter combinations inherent in statistical sampling. This efficiency advantage becomes especially important in multi-layer resistivity problems where the parameter space remains manageable, and the electromagnetic wave propagation through layered media is well-understood through Maxwell's equations and mathematical solutions.

Understanding Middle East Carbonate Reservoirs

Middle East carbonate reservoirs, containing the world's largest oil accumulations, display unique electrical characteristics that challenge traditional resistivity interpretation methods developed for clastic sandstone formations. Recognition of these distinctive features is essential for successful electromagnetic method application in these economically significant environments that continue to supply a major portion of global oil production.

The defining characteristic of Middle East carbonates is their inverse resistivity behavior compared to conventional sandstone reservoirs. While oil-bearing sandstones typically exhibit high resistivity readings, Middle East carbonates often display surprisingly relatively lower resistivity within the hydrocarbon bearing formations and its intercalations. This phenomenon occurs because of the intricate pore structure typical of carbonate rocks, where oil preferentially fills large vugs, caverns, and fractures while conductive formation water stays trapped in tiny micropores within the rock matrix. This dual-porosity system creates parallel conductive paths through the water-wet microporosity that substantially lowers the overall formation resistivity, contradicting traditional interpretation approaches developed for water-wet sandstone systems.

The complex pore structure in carbonates results from various diagenetic processes including dissolution by meteoric waters or hydrothermal fluids, cementation during burial, dolomitization, and tectonic fracturing. Primary porosity from original depositional textures, whether grainstone, packstone, or wackestone fabrics, often undergoes significant modification through these secondary processes. The resulting pore systems may range from submicron micropores to centimeter-scale vugs and caverns, creating a heterogeneous distribution that significantly affects fluid distribution and electrical properties.

Non-reservoir sections consisting of tight carbonates with minimal porosity and permeability exhibit high resistivity due to the absence of conductive fluid pathways. These tight zones often consist of extensively cemented limestone or crystalline dolomite where diagenetic processes have occluded pore space. This creates conditions where productive zones may show lower resistivity than the surrounding tight rocks, complicating the interpretation and requiring careful calibration with additional petrophysical measurements including nuclear magnetic resonance, formation testing, and borehole imaging.

The extensive production history of these reservoirs introduces additional complexity to resistivity interpretation. Many major fields have been producing since initial discoveries in the 1930s-1950s, with peak production periods in the 1970s-1980s, during which operators have implemented various enhanced oil recovery methods. Water flooding, initiated in many fields during the 1960s-1970s, has created complex saturation distributions. Miscible gas injection, polymer flooding, and chemical recovery techniques have further altered original fluid distributions. These processes have created saturation patterns with transition zones spanning hundreds of feet vertically and extending laterally for kilometers. Traditional static interpretation models based on virgin reservoir conditions, assuming clean oil-water contacts at structural spill points, are insufficient for current field evaluation.

Long production history combined with enhanced recovery operations create substantial uncertainty in determining current oil-water boundaries. Water breakthrough from injection wells produces irregular flood fronts influenced by reservoir heterogeneity, creating fingering and channeling effects that resist simple geometric modeling. Gravity segregation in thick reservoir intervals and permeability variations result in variable saturation profiles that deviate significantly from original structural boundaries. Injected fluids preferentially follow high-permeability streaks, fracture networks, and karst features, creating complex patterns that bypass significant oil volumes in lower permeability matrix rock. The inherently small resistivity differences between oil and water zones, make precise fluid contact identification through particularly challenging electromagnetic methods.

Complex reservoir architecture in Middle East carbonates necessitates sophisticated well trajectories to maximize reservoir contact and optimize production. Multiple transitions between reservoir layers separated by dense barriers such as stylolite, hardgrounds, or anhydrite stringers require precise trajectory

control. Single horizontal wellbores accessing multiple reservoir compartments replace multiple vertical wells, significantly reducing development costs and environmental footprint. However, structural features including faults, tight-radius folds, and karst collapse features may cause unexpected exits from the target zones. Azimuthal resistivity measurements combined with real-time analysis enable immediate decision-making for trajectory correction, preventing costly sidetracks and ensuring optimal reservoir exposure.

Applying the Method to Carbonate Systems

Implementation of multi-layer azimuthal resistivity inversion in Middle East carbonate reservoirs provides substantially improved performance compared to traditional distance-to-boundary methods in low-resistivity contrast environments characteristic of these formations. The deterministic algorithm specifically addresses these challenging conditions through robust mathematical foundations, and flexible parameterization adapting to geological complexity.

The multi-layer inversion approach provides comprehensive characterization of both structural arrangement and fluid saturation profiles in carbonate reservoirs through analysis of formation resistivity at multiple investigation depths. Under favorable measurement conditions with adequate tool spacing and frequency range, the technique demonstrates the ability to find reliable solutions with enhanced radial depths of investigation while simultaneously determining formation thickness and resistivity values across multiple layers. This capability proves to be particularly valuable where structural complexity from folding, faulting, and stratigraphic variations combines with fluid distribution uncertainty from decades of production.

The algorithm effectively handles minimal resistivity differences through advanced regularization techniques and constraint application based on geological understanding. By maintaining numerous adjustable parameters in the inversion process while applying physically reasonable constraints, the method can distinguish between subtle resistivity variations that would be impossible to resolve using conventional techniques. This capability is particularly valuable in mature fields where enhanced recovery operations have created complex three-dimensional fluid distributions requiring detailed characterization for production optimization and infill drilling planning.

The application of the method in real-time during drilling operations enables immediate interpretation of formation properties and fluid content, supporting wellbore placement decisions within narrow target windows often less than 10 feet thick. The inversion processing typically completes within 1-2 seconds per measurement point along the well trajectory, enabling continuous model updates as new data becomes available. When unexpected geological features are encountered, such as sub-seismic faults, diagenetic boundaries, or unexpected fluid contacts, azimuthal resistivity inversion provides quantitative information about surrounding formation structure, supporting informed steering decisions. Resolution of multiple layers simultaneously provides both local detail around the wellbore and broader geological context, essential for navigation through complex carbonate sequences where lateral facies changes can occur.

The integration process involves continuous data acquisition from logging-while-drilling tools, automated quality control to identify and filter noise or tool artifacts, real-time inversion processing, continuously monitored by the user, typically completing within seconds of data acquisition, and immediate visualization of results for drilling decision support. This workflow ensures that geological insights remain available throughout the drilling process, enabling proactive rather than reactive well placement strategies.

Integration of this inversion methodology with logging-while-drilling tools represents a fundamental change in carbonate reservoir characterization capabilities. Reliable multi-layer models in real-time, even in challenging low-contrast environments, enable operators to optimize well placement with respect to fluid contacts and reservoir boundaries, minimize drilling risks associated with unexpected exits from target zones, and improve recovery from complex reservoir systems through better understanding of reservoir architecture. Successful application in Middle East carbonates demonstrates potential for deployment in other challenging geological environments.

Algorithm Validation and Testing

Before deploying any new interpretation method in field operations, rigorous validation becomes essential to ensure both accuracy and reliability. Extensive testing of the multi-layer inversion algorithm has been conducted using the industry-standard SPWLA (Society of Petrophysicists and Well Log Analysts) Resistivity SIG (Special Interest Group) benchmark models, which provide standardized scenarios for evaluating electromagnetic interpretation techniques.

The SDAR (Standard Deep Azimuthal Resistivity) Workgroup developed benchmark models specifically to challenge interpretation algorithms and verify performance across various complexity levels ranging from simple two-layer cases to sophisticated multi-layer scenarios. The benchmark suite enables objective comparison between different commercial tools and algorithms through standardized test conditions, ensuring consistent performance evaluation across the industry. Testing of the proposed method focused particularly on the seven-layer model D1, which presents a sophisticated resistivity structure that closely resembles the complex layering encountered in carbonate reservoir environments. The configuration tests the algorithm's ability to resolve thin layers and handle various resistivity contrasts.

The seven-layer D1 model incorporates subtle resistivity contrasts and varying thickness sequences characteristic of Middle East carbonates where diagenetic processes create complex porosity distributions. The algorithm successfully resolved all seven layers with their correct resistivity values and thickness, demonstrating its capability to handle complex multi-layer scenarios. Results demonstrated excellent agreement between inverted parameters and known model properties (Figure 1). In the following 2 figures it is possible to appreciate the results of the inversion when using 2 opposite approaches: The 100% unconstrained where it is assumed that there is no knowledge of the geometry and resistivity of the layers, and the constrained, where a 10% range window has been applied around the true values proposed for each parameter.

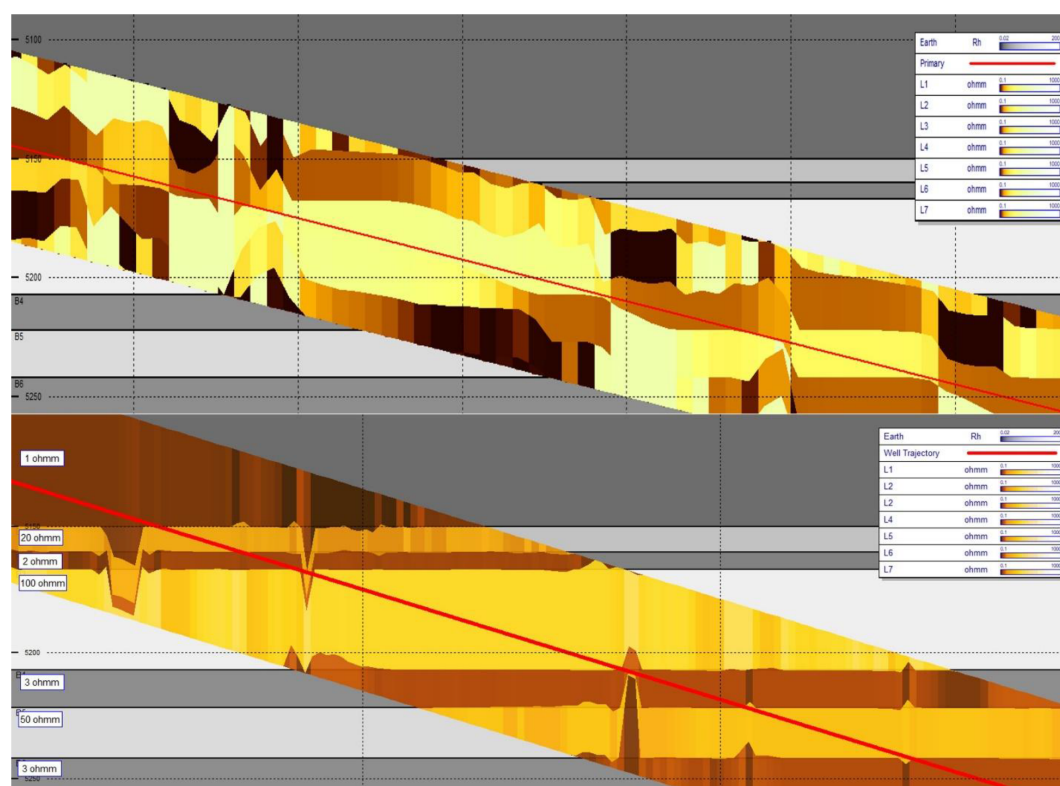


Figure 1—Unconstrained vs. tightly constrained inversion on the 7 Layers Model D1

Following the benchmark validation, the testing proceeded using data from previously drilled wells where conventional two-boundary inversion had been applied during original drilling campaigns. This real-data validation included wells from different fields, various reservoir intervals, and diverse drilling conditions, providing comprehensive assessment of method robustness. This real-data validation provided crucial insights into the practical advantages of multi-layer analysis over traditional approaches.

The multi-layer inversion improved the interpretation capability and geological insight impacting directly drilling decisions and reservoir understanding. It identifies geological features beyond the nearest boundaries that remain undetected by methods limited to three layers / two boundaries setup.

Traditional two-boundaries algorithms, while computationally efficient and widely implemented in commercial software, inherently restrict the interpretation. Important geological features such as thin beds, internal reservoir zonation, or major boundaries beyond the immediate thin screening layer remained invisible to the interpretation.

Within high-contrast boundaries identified as single interfaces by traditional inversion, the multi-layer approach detects internal zonation related to various geological factors. Diagenetic development creates zones with different porosity types and permeability characteristics. Varying pore structures from microporous mudstone to vuggy grainstone affect fluid distribution and production potential. Fluid saturation patterns from production history create complex resistivity profiles requiring detailed characterization. This internal structure provides critical information for understanding reservoir heterogeneity, predicting production performance, and supporting well placement decisions to maximize hydrocarbon recovery.

Enhanced resolution through systematic parameter space exploration enables improved detection of layer boundaries and interfaces within complex carbonate sequences where rapid facies changes occur over distances of tens to hundreds of feet. Rather than interpreting these zones as single boundaries with average properties that obscure important details, the multi-layer approach resolves constituent layers with their individual properties, providing detailed structural information about formation architecture including thickness variations, lateral extent, and continuity (Figure 2). This detailed characterization proves essential for understanding reservoir compartmentalization, identifying flow barriers, and optimizing completion strategies.

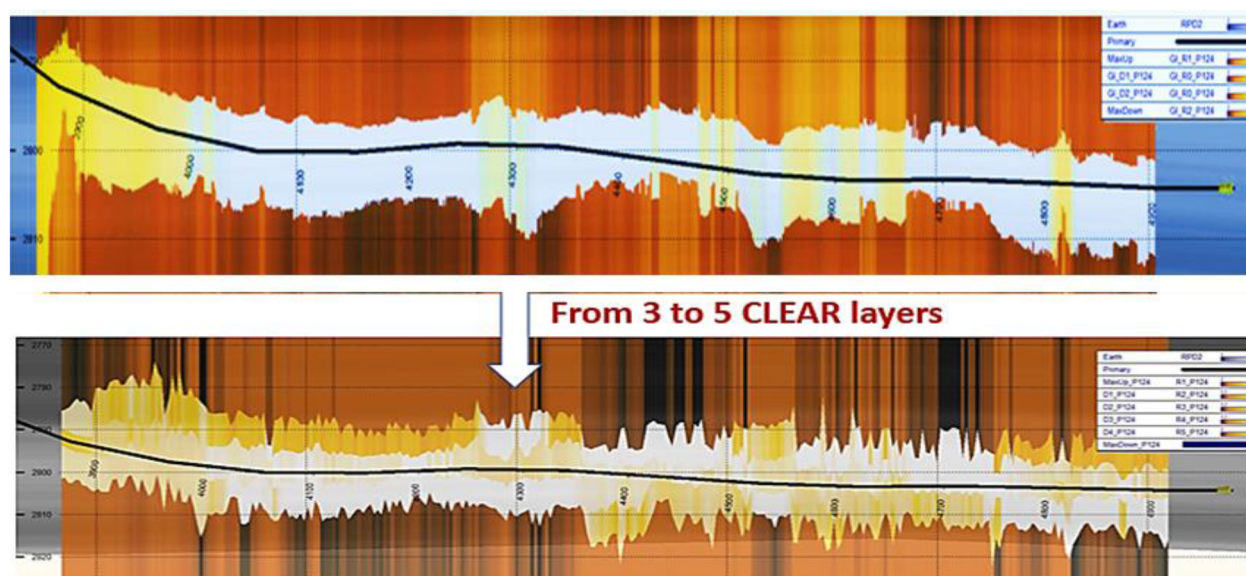


Figure 2—Real time results of standard 2 boundaries inversion vs. multi-layer inversion interpretation

Field Applications in Middle East Operations

Deployment of the multi-layer azimuthal resistivity inversion algorithm in multiple Middle East carbonate drilling campaigns has provided extensive real-world validation of method capabilities and demonstrated significant operational value in diverse geological settings.

In the first case study (Figure 3), multi-layer inversion tracked zonation within a thick reservoir layer previously interpreted as homogeneous. Analysis revealed distinct zones of higher resistivity correlating with reduced reservoir quality, attributed to cementation and porosity reduction. This detailed characterization of resistivity variations within the reservoir interval helped the drilling team understand local geological structure influenced by differential diagenesis and supported trajectory adjustment decisions to maintain the wellbore within optimal porosity zones while avoiding tight streaks.

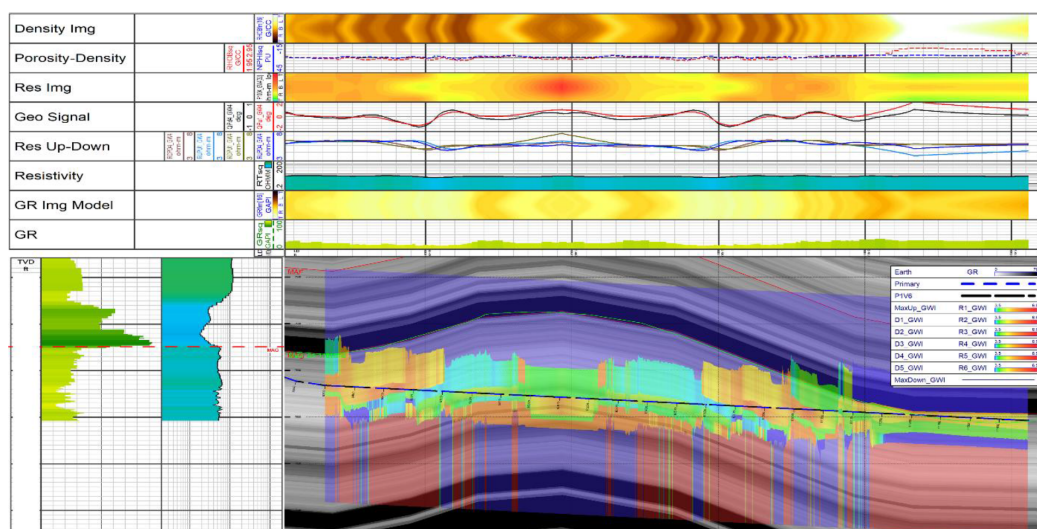


Figure 3—Mapping reservoir zonation across localized structural height

The second application demonstrated improved precision in mapping reservoir boundaries and overlying seal integrity (Figure 4). The algorithm determined thickness variations of the overlying dense formation along the wellbore trajectory. This information proved essential for seal integrity analysis, indicating areas where the seal might be compromised by faulting or erosion, and supporting completion design optimization including isolation packer placement and perforation strategy.

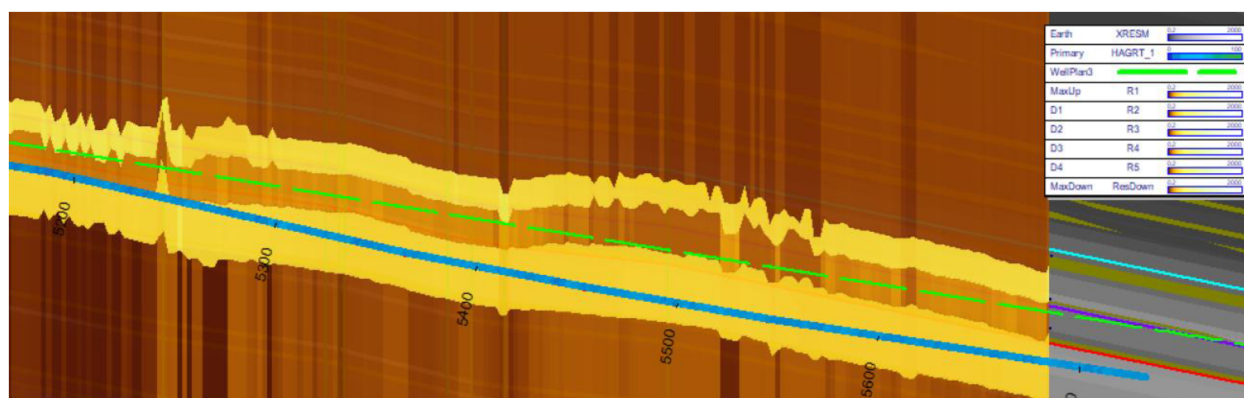


Figure 4—Mapping thickness changes of sealing layer

The third implementation involved an extremely extended reach drilling operation exceeding 18,000 feet measured depth with trajectory optimization between two thin reservoir layers each less than 5 feet thick (Figure 5).

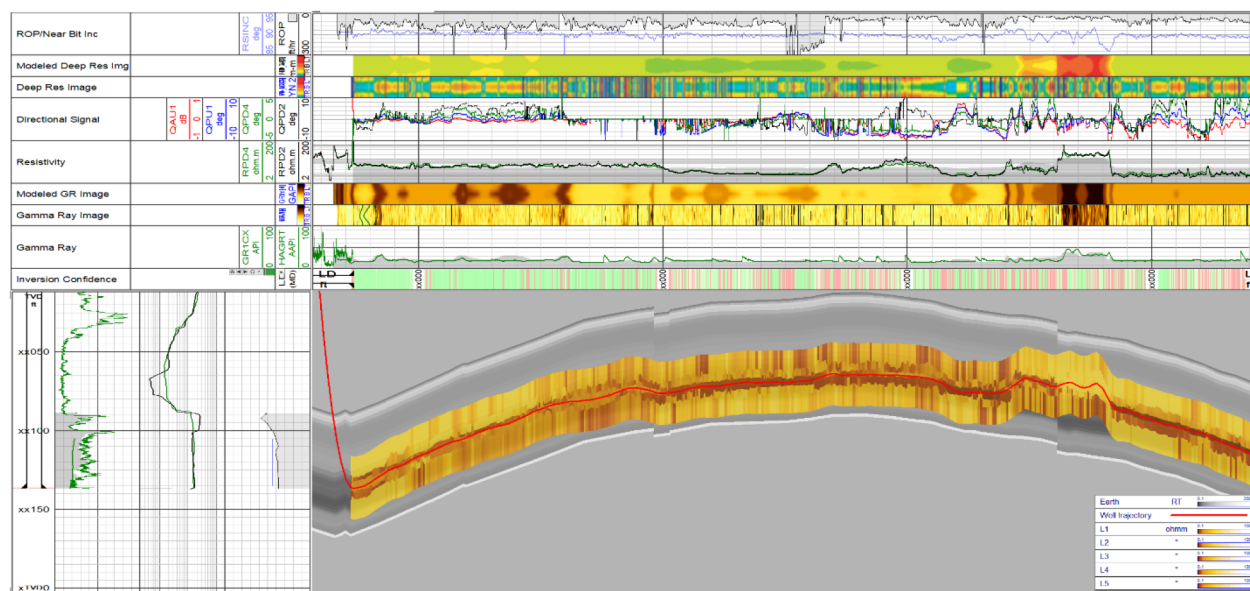


Figure 5—Extended reach well with moving between the targets

Multi-layer inversion helped to optimize the trajectory by minimizing exposure to non-productive intervals while maintaining dogleg severity within tight limits. The algorithm provided continuous structural evaluation during drilling, helping to optimize entry and exit points for layer transitions. During fault zone penetration, inversion results enabled efficient trajectory planning for target zone re-entry, avoiding an expensive sidetrack and saving rig time (Figure 6).

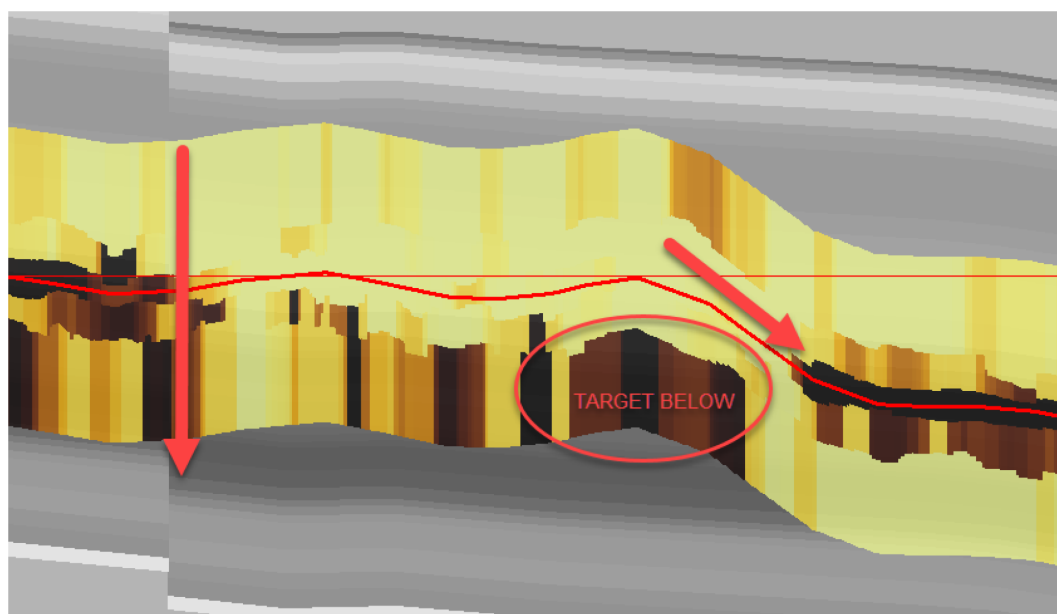


Figure 6—Re-entry in the target after a throw-down fault

Field experiences across different geological settings demonstrate that multi-layer azimuthal resistivity inversion provides actionable geological information directly supporting drilling decisions and enhancing

reservoir understanding. Real-time processing capabilities enable immediate access to enhanced structural information during drilling operations, contributing to informed decision-making in carbonate environments where precise trajectory control and accurate formation characterization are essential for operational success and economic optimization.

Conclusion

Development and successful implementation of multi-layer azimuthal resistivity inversion represents a significant practical advancement in real-time formation evaluation for complex carbonate reservoir environments where traditional methods have proven inadequate. The Newton-based non-linear one-dimensional inversion algorithm achieves high computational efficiency of approximately 1.5 seconds per calculated measured depth point while maintaining consistent performance during fast drilling operations. This processing capability ensures continuous availability of geological information to drilling teams regardless of operational tempo or data acquisition rates.

The integrated deep search algorithm optimizes parameter estimation through systematic solution space exploration using gradient-based methods, identifying reliable resistivity and thickness values for each layer in the formation model. This optimization process enhances result robustness when dealing with subtle resistivity contrasts characteristic of Middle East carbonate reservoirs, where resistivity ratios between layers may be very low. Algorithm flexibility accommodates formation structures from simple two-layer models representing single boundary conditions, possibly for landing optimization applications, to complex ten-layer configurations incorporating multiple reservoir and non-reservoir intervals, providing versatility for various geological scenarios encountered in heterogeneous carbonate sequences.

Quality control through comprehensive misfit analysis automatically evaluates solution confidence and identifies interpretation uncertainties arising from measurement noise, parameter correlation, or model non-uniqueness. This assessment capability enables real-time result validation and supports risk-based decision-making during drilling operations where trajectory adjustments may bear costly economic implications. The misfit analysis provides quantitative feedback about data quality and solution stability - essential information for operational decisions.

This multi-layer inversion approach functions as an integral component of comprehensive geological evaluation rather than a standalone interpretation tool. Enhanced structural and resistivity information complements other formation evaluation measurements, contributing to integrated reservoir characterization and well placement planning. This holistic approach ensures optimal utilization of all available data for reservoir understanding.

Validation through benchmark testing, achieving consistent accuracy across standardized test cases and successful field applications in Middle East carbonate reservoirs with diverse geological characteristics demonstrates practical applicability of this interpretation methodology. Field implementations, world-wide, have successfully characterized internal reservoir zonation, mapped boundaries, and optimized complex well trajectories including extended reach operations. The algorithm proved also valuable during fault crossing operations, where accurate formation characterization enabled efficient trajectory planning and successful target zone re-entry.

By providing detailed real-time insights into complex formation structures, multi-layer inversion enables informed drilling decisions and contributes to improved understanding of carbonate reservoir systems containing remaining oil reserves. The demonstrated capability to support challenging operations while maintaining performance under demanding conditions positions this methodology as a valuable tool for carbonate reservoir development in the Middle East and similar geological environments worldwide. The approach represents an evolutionary improvement that enhances utilization of existing azimuthal resistivity tools through advanced yet computationally efficient processing techniques, delivering practical benefits for drilling optimization and reservoir characterization.

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